Introduction to Semantic Analysis

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The Compiler so far

- Lexical Analysis
  - Detects inputs with illegal tokens
- Parsing
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- Lexical Analysis
  - Detects inputs with illegal tokens
- Parsing
  - Detects inputs with ill-formed parse trees
- Next: Semantic Analysis
  - Last ‘front-end’ phase
  - Catches all the remaining errors

Semantic Analysis

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Semantic routines:

- interpret (partial) meaning of the program based on its syntactic structure
- two purposes:
  - finish analysis by deriving context-sensitive information (e.g. type checking)
  - begin synthesis by generating the IR or target code

What does Semantic Analysis do?

- Checks of many kinds:
  - All identifiers are declared.
  - Type checking in assignment statements, function calls, expressions, etc.
  - Inheritance relationships.
  - Classes defined only once.
  - Methods in a class defined only once.
  - Reserved identifiers are not misused

  And many others...

Why a Separate Semantic Analysis?

Parsing cannot catch some errors. Some language constructs are not context-free.

- All identifiers are declared before use.
  - Corresponds to the abstract language $\{wcw \mid w, c \in \{a,b\}^*\}$.
  - The first w represents an identifier declaration; the second w represents a use of the same identifier.

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- The number of formal parameters in the declaration of a function agrees with the number of actual parameters in the use of the function.
  - Corresponds to the abstract language $\{a^n b^m c^n d^m\}$.
  - Here, a^n, b^m represent the formal-parameter list of two functions, while c^n, d^m represent the actual-parameter list in calls to the functions.

These languages are not context-free.

Scope

- Matching identifier declarations with uses
  - One of the most important tasks performed by Semantic Analysis.
- The scope of an identifier is the portion of a program in which that identifier is accessible.
- The same identifier may refer to different things in different parts of the program.
  - Different scopes for same name don't overlap.
- An identifer may have restricted scope even if declared before use.

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   y = x;
   int x = 1;
   z = x;
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Uses of `x` refer to closest enclosing definition.

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- A dynamically scoped variable refers to the closest enclosing binding in the execution of the program.
  - Lisp, SNOBOL have dynamic scoping for variables.
- The exception handler for an exception is dynamically determined.
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Example of a language with dynamic scoping for variables:

```
let a = 0;
f() = a;
g(y) = let a = 4 in f();
h(z) = let a = 5 in f();
```

Symbol tables

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- variable names
- defined constants
- procedure and function names
- source text labels
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A symbol table is a compile-time structure

Symbol table information

What kind of information might the compiler need?

- textual name
- data type
- declaring procedure
- lexical level of declaration
- storage class (base address)
- offset in storage
- can it be aliased? to what other names?
- number and type of arguments to functions
- …

Storage classes of variables

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The storage classes also determine how the variables are addressed (addressing mode).

- A local variable is not assigned a fixed machine address (or relative to the base of a module)
- Rather a stack location that is accessed by an offset from a register whose value may be different every time the procedure is invoked.

Symbol table organization

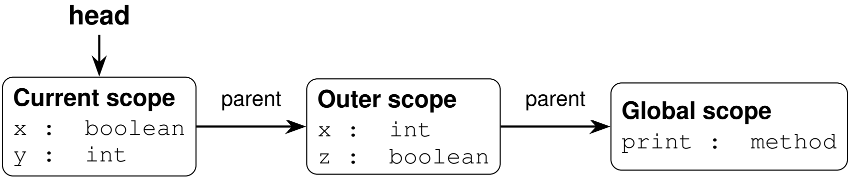
How should the table be organized?

- Linear List
  - **O**(n) probes per lookup
  - easy to expand — no fixed size
  - one allocation per insertion
- Ordered Array
  - **O**($\log_2 n$) probes per lookup using binary search
  - insertion is expensive (to reorganize list)
- Binary Search Tree
  - **O**(n) probes per lookup — unbalanced
  - **O**($\log_2 n$) probes per lookup — balanced
  - easy to expand — no fixed size
  - one allocation per insertion
- Hash Table
  - **O**(1) probes per lookup — on average
  - expansion costs vary with specific scheme

Nested scopes: block-structured symbol tables

What information is needed?

- when asking about a name, want <u>most recent</u> declaration
- declaration may be from current scope or outer scope
- innermost scope overrides outer scope declarations



lookup proceeds from the head outward

Key point: new declarations occur only in current scope

Nested scopes: block-structured symbol tables

What operations do we need?

- `void put (Symbol key, Object value)`
  bind key to value
- `Object get(Symbol key)`
  return value bound to key
- `void beginScope()`
  open a new scope and remember current state of table
- `void endScope()`
  close current scope and restore table to state at most recent open beginScope

Attribute information

Attributes are internal representation of declarations

Symbol table associates names with attributes

Names may have different attributes depending on their meaning:

- variables: type, procedure level, frame offset
- types: type descriptor, data size/alignment
- constants: type, value
- procedures: formals (names/types), result type, block information (local decls.), frame size

Types

- What is a type?
  - A set of values.
  - A set of operations on those values.
- Classes (e.g. in Java) are one instantiation of the modern notion of type.

Type system

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Why do we need Type Systems?

- Not all operations are legal for all types.
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- But both have the same assembly language implementation: `add $r1, $r2, $r3`.

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A language's type system specifies which operations are valid for which types.

- The goal of *Type Checking* is to ensure that operations are used with correct types.

Type Checking Overview

- Three kinds of languages:
  - *Statically typed*: All or almost all checking of types is done as part of compilation (C, Java, OCaml, Rust, MiniJava).
  - *Dynamically typed*: Almost all checking of types is done as part of program execution (Python, JavaScript, Scheme)
  - *Untyped*: No type checking (machine code)

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- Recent trend: bolt-on static type checking on top of dynamically typed languages.
  - See Hack (Meta, PHP), Flow (Meta, JavaScript), TypeScript (Microsoft, JavaScript), Eqwalizer (WhatsApp, Erlang), Pyre (Meta, Python).

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This is a type war. There is no clear winner.

Type Checking and Type Inference

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- The formalism to express both type-checking and type-inference is *logical rules of inference*.
  - Just like how the formal notation for lexical analysis and parsing were regular expressions and context-free grammars respectively.

Rules of Inference

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  If Hypothesis is true, then Conclusion is true.
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- Inference rules are often written as follows:

$$\frac{\vdash \text{Hypothesis}_1 \quad \vdash \text{Hypothesis}_2 \ \ldots \ \vdash \text{Hypothesis}_n}{\vdash \text{Conclusion}}$$

This translates to: **If** Hypothesis$_1$ **and** Hypothesis$_2$ **and** $\ldots$Hypothesis$_n$ **then** Conclusion.

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Inference rules are a very succinct way to precisely state type-checking rules.

Rules of Inference

$$\frac{\vdash E1 : T1 \quad \vdash E2 : T2}{\vdash E3 : T3}$$

- For type inference, the rule is read as:
  - *If $E1$ has type $T1$ and $E2$ has type $T2$ then $E3$ has type $T3$.*
- For type checking, the rule is read as:
  - *If E_1 with type $T1$ and E_2 with type $T2$ type-check, then E_3 also type-checks with type $T3$.*
  - We may sometimes omit $T3$ when type checking (statements, for example).

Examples

$$\frac{\text{program contains } \texttt{int i}}{\vdash \texttt{i} : \mathit{int}}$$

$$\frac{\vdash e_1 : \mathit{int} \quad \vdash e_2 : \mathit{int}}{\vdash e_1 + e_2 : \mathit{int}}$$

$$\frac{\vdash \mathit{id} : \mathit{int} \quad \vdash e : \mathit{int}}{\vdash \mathit{id} = e}$$

Type Environment

- A type environment is a mapping from identifiers to types.
  - We will use the symbol A to denote type environment.
  - $dom(A)$ denotes the domain of A, i.e. the set of identifiers for which types are defined in the environment A.
  - We use the notation $A \vdash id : T$ to denote that $A(id) = T$.

Type expressions

Type expressions are a textual representation for types:

1. basic types: $boolean$, $char$, $integer$, $real$, etc.
2. type names
3. constructed types (constructors applied to type expressions):
    1. $array(I, T)$ denotes an array of T indexed over I
       e.g., $array(1 \ldots 10, integer)$
    2. products: $T_1 \times T_2$ denotes Cartesian product of type expressions T_1 and T_2
    3. records: fields have names
       e.g., $record((\mathtt{a} \times integer), (\mathtt{b} \times real))$
    4. pointers: $pointer(T)$ denotes the type "pointer to an object of type T"
    5. functions: $D \rightarrow R$ denotes the type of a function mapping domain type D to range type R
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In Minijava, the type expressions are `boolean`, `int`, `int[]`, id.

MiniJava Type Judgements

$\vdash g$		The goal g type checks
$\vdash mc$		The main class mc type checks
$\vdash d$		The type declaration d type checks
$C \vdash m$		If defined under class C, the method m type checks
$A, C \vdash s$		In a type environment A, if written in class C, the statement s type checks
$A, C \vdash e : t$		In a type environment A, if written in class C, the expression e has type t
$A, C \vdash p : t$		In a type environment A, if written in class C, the primary expression p has type t

Typechecking a Minijava program

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Our goal is to typecheck all the classes:

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Our goal is to typecheck all the classes:

$$\frac{distinct(classname(mc), classname(d_1), \ldots, classname(d_n)) \qquad \vdash mc \quad \vdash d_i, i \in \{1, \ldots, n\}}{\vdash mc\ d_1 \ldots d_n}$$

This says that if all the class-names are distinct, and if all the classes individually type-check, then the entire program also type-checks.

Ref: The MiniJava Type System, Jens Palsberg.

$$\frac{\begin{array}{c} \mathit{distinct}(id_1, \ldots, id_f) \\ \mathit{distinct}(\mathit{methodname}(m_1), \ldots, \mathit{methodname}(m_k)) \\ id \vdash m_i \quad i \in \{1, \ldots, k\} \end{array}}{\vdash \texttt{class}\ \ id\{t_1\ id_1;\ \ldots;\ t_f\ id_f;\ m_1\ \ldots m_k\}}$$

$id \vdash m_i$ means that method m_i type-checks when defined in class id. The method rule constructs its type environment from the fields of class id, the method's formal parameters, and its local variables.

Typechecking a Minijava class

$$\frac{\begin{array}{c} distinct(id_1, \ldots, id_f) \\ distinct(methodname(m_1), \ldots, methodname(m_k)) \\ id \vdash m_i \quad i \in \{1, \ldots, k\} \end{array}}{\vdash \texttt{class} \;\; id\{t_1 \; id_1; \; \ldots; \; t_f \; id_f; \; m_1 \; \ldots m_k\}}$$

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Question: Does this allow a method and a field to have the same name?

Ref: The MiniJava Type System, Jens Palsberg.

“fields” helper function

$$\frac{\texttt{class}\ \ id\ \{t_1\ id_1;\ \ldots;\ t_f\ id_f;\ m_1\ \ldots\ m_k\}\ \text{is in the program}}{\mathit{fields}(id) = [id_1 : t_1, \ldots, id_f : t_f]}$$

$$\frac{\texttt{class}\ \ id\ \texttt{extends}\ \ id^P\ \{t_1\ id_1;\ \ldots;\ t_f\ id_f;\ m_1\ \ldots\ m_k\}\ \text{is in the program}}{\mathit{fields}(id) = \mathit{fields}(id^P) \cdot [id_1 : t_1, \ldots, id_f : t_f]}$$

Scope convention: In $A \cdot B$, the right-hand side B is the inner scope. Thus, a binding in B shadows one with the same name in A.

Typechecking a Minijava method

$$\frac{\begin{array}{c} distinct(id_1^F, \ldots, id_n^F) \\ distinct(id_1, \ldots, id_r) \\ A = fields(C) \cdot [id_1^F : t_1^F, \ldots id_n^F : t_n^F] \cdot [id_1 : t_1, \ldots id_r : t_r] \\ A, C \vdash s_i,\ i \in \{1, \ldots, q\} \quad A, C \vdash e : t' \quad t' \leq t \end{array}}{\begin{array}{c} C \vdash \texttt{public}\ \ t\ id^M\ (t_1^F\ id_1^F,\ \ldots,\ t_n^F\ id_n^F)\{ \\ t_1\ id_1;\ \ldots;\ t_r\ id_r;\ s_1\ \ldots s_q;\ \texttt{return}\ \ e;\} \end{array}}$$

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- Does this allow a local variable and a field to have the same name?

Ref: The MiniJava Type System, Jens Palsberg.

Inheritance in Minijava

To express inheritance among classes, we will define the *subtype* relation, denoted by $\leq$.

For class types, $t_1 \leq t_2$ iff $t_1 = t_2$ or t_1 is a subclass of t_2.
Reflexivity also includes `int`, `boolean`, and `int[]`.

The following rules describe the *subtype* relation:

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$$t \leq t$$

$$\frac{t_1 \leq t_2 \quad t_2 \leq t_3}{t_1 \leq t_3}$$

$$\frac{\texttt{class}\ C\ \texttt{extends}\ D \quad \text{is in the program}}{C \leq D}$$

Typechecking statements in Minijava

$$\frac{A(id) = t_1 \qquad A, C \vdash e : t_2 \qquad t_2 \leq t_1}{A, C \vdash id \ \texttt{=}\ e\texttt{;}}$$

$$\frac{A(id) = \texttt{int[]} \qquad A, C \vdash e_1 : \texttt{int} \qquad A, C \vdash e_2 : \texttt{int}}{A, C \vdash id\ \texttt{[}\ e_1\ \texttt{]}\ \texttt{=}\ e_2\texttt{;}}$$

$$\frac{A, C \vdash e : \texttt{boolean} \qquad A, C \vdash s_1 \qquad A, C \vdash s_2}{A, C \vdash \texttt{if}\ \texttt{(}\ e\ \texttt{)}\ s_1\ \texttt{else}\ s_2}$$

$$\frac{A, C \vdash e : \texttt{boolean} \qquad A, C \vdash s}{A, C \vdash \texttt{while}\ \texttt{(}\ e\ \texttt{)}\ s}$$

$$\frac{A, C \vdash e : \texttt{int}}{A, C \vdash \texttt{System.out.println(}\ e\ \texttt{);}}$$

Ref: The MiniJava Type System, Jens Palsberg.

Typechecking expressions in Minijava

$$\frac{A, C \vdash p_1 : \texttt{boolean} \qquad A, C \vdash p_2 : \texttt{boolean}}{A, C \vdash p_1 \ \texttt{\&\&}\ p_2 : \texttt{boolean}}$$

$$\frac{A, C \vdash p_1 : \texttt{int} \qquad A, C \vdash p_2 : \texttt{int}}{A, C \vdash p_1 \ \texttt{<}\ p_2 : \texttt{boolean}}$$

$$\frac{A, C \vdash p_1 : \texttt{int} \qquad A, C \vdash p_2 : \texttt{int}}{A, C \vdash p_1 \ \texttt{+}\ p_2 : \texttt{int}}$$

$$\frac{A, C \vdash p_1 : \texttt{int} \qquad A, C \vdash p_2 : \texttt{int}}{A, C \vdash p_1 \ \texttt{-}\ p_2 : \texttt{int}}$$

$$\frac{A, C \vdash p_1 : \texttt{int} \qquad A, C \vdash p_2 : \texttt{int}}{A, C \vdash p_1 \ \texttt{*}\ p_2 : \texttt{int}}$$

$$\frac{A, C \vdash p_1 : \texttt{int[]} \qquad A, C \vdash p_2 : \texttt{int}}{A, C \vdash p_1 \ \texttt{[}\ p_2 \ \texttt{]} : \texttt{int}}$$

$$\frac{A, C \vdash p : \texttt{int[]}}{A, C \vdash p \ \texttt{.length} : \texttt{int}}$$

Typechecking method calls in Minijava

$$\frac{???}{A, C \vdash p.id(e_1, e_2, \ldots, e_n) : t}$$

Typechecking method calls in Minijava

$$\frac{A,C \vdash p : D \quad methodtype(D,id) = (t_1^{'},\ldots,t_n^{'}) \rightarrow t \qquad A,C \vdash e_i : t_i \quad t_i \leq t_i^{'},\ i \in \{1,\ldots,n\}}{A,C \vdash p.id(e_1,e_2,\ldots,e_n) : t}$$

$methodtype(D,id)$ returns the type of the method id in class D or one of its super-classes.

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Question: What change should we make in the rule if we want to enforce access modifiers?

Typechecking method calls in Minijava with access modifiers

$$\frac{\begin{array}{c} A, C \vdash p : D \quad methodtype(D, id) = (t_1^{'}, \ldots, t_n^{'}) \rightarrow t \\ (methodaccess(D, id) = \texttt{public} \\ \vee\ methodaccess(D, id) = \texttt{protected} \wedge\ C \leq D \\ \vee\ methodaccess(D, id) = \texttt{private} \wedge\ C = D) \\ A, C \vdash e_i : t_i \quad t_i \leq t_i^{'}\ i \in \{1, \ldots, n\} \end{array}}{A, C \vdash p.id\ (e_1, e_2, \ldots, e_n) : t}$$

$methodtype(D, id)$ returns the type of the method id in class D or one of its super-classes.

$methodaccess(D, id)$ returns the access modifier of the method id in class D or one of its super-classes.

Typechecking ternary operation in Minijava

$$\frac{???}{A, C \vdash (e_1 ? e_2 : e_3) : t}$$

Typechecking ternary operation in Minijava

$$\frac{A, C \vdash e_1 : \texttt{boolean} \quad A, C \vdash e_2 : t \quad A, C \vdash e_3 : t}{A, C \vdash (e_1 ? e_2 : e_3) : t}$$

Typechecking ternary operation in Minijava

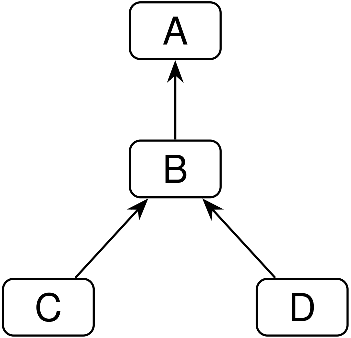
$$\frac{A, C \vdash e_1 : \texttt{boolean} \quad A, C \vdash e_2 : t \quad A, C \vdash e_3 : t}{A, C \vdash (e_1 ? e_2 : e_3) : t}$$

But what about inheritance? How to handle subtyping?

Example

```
...
class B extends A {...}
class C extends B {...}
class D extends B {...}
...
A objA; B objB; C objC; D objD;
...
objA = (...) ? objC : objD;
objB = (...) ? objC : objD;
```

Class hierarchy



Arrows point to the superclass.

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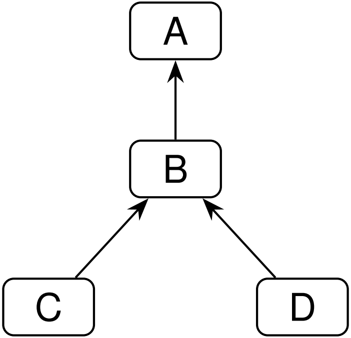

Arrows point to the superclass.

Both the instances of ternary statement should type-check. This requires the ternary statement itself to have the type `B`.

Least Upper Bounds

- $lub(C,D)$, the least upper bound of classes C and D is defined to class E such that
  1. $C \leq E$ and $D \leq E$: E is an upper bound of both C and D
  2. $\forall E'.\, C \leq E' \wedge D \leq E' \Rightarrow E \leq E'$: E is least among upper bounds.
- In Minijava, the least upper bound of two classes is their least common ancestor in the inheritance tree.

Typechecking ternary operation in Minijava: Attempt-2

$$\frac{A, C \vdash e_1 : \texttt{boolean} \quad A, C \vdash e_2 : t_1 \quad A, C \vdash e_3 : t_2 \qquad \exists t'.\ t_1 \leq t' \wedge t_2 \leq t' \quad t = lub(t_1, t_2)}{A, C \vdash (e_1 ? e_2 : e_3) : t}$$

Typechecking No-overloading in Minijava

- **Overloading:** A method is said to be overloaded if there exists another method in the same class or its super classes with the same name but different parameter types or parameter count.
- **Overriding:** A method in a class is said to be overridden if a subclass declares a method with the same name and parameter types, and the same or a subtype of the inherited return type.

$$\frac{\begin{array}{c} distinct(id_1, \ldots, id_f) \\ distinct(methodname(m_1), \ldots, methodname(m_k)) \\ id \vdash m_i,\ i \in \{1, \ldots, k\} \\ ??? \end{array}}{\vdash \texttt{class}\ id\ \texttt{extends}\ id^P\{t_1\ id_1;\ \ldots;\ t_f\ id_f;\ m_1\ \ldots m_k\}}$$

Typechecking No-overloading in Minijava

$$\frac{\begin{array}{c} distinct(id_1,\ldots,id_f) \\ distinct(methodname(m_1),\ldots,methodname(m_k)) \\ id \vdash m_i,\ i \in \{1,\ldots,k\} \\ noOverloading(id, id^P, methodname(m_i)),\ i \in \{1,\ldots,k\} \end{array}}{\vdash \texttt{class}\ id\ \texttt{extends}\ id^P\{t_1\ id_1;\ \ldots;\ t_f\ id_f;\ m_1\ \ldots m_k\}}$$

$noOverloading(C,P,m)$ holds exactly when:

- $methodtype(P,m) = \perp$; or
- $methodtype(C,m) = (t_1,\ldots,t_n) \to t$,
  $methodtype(P,m) = (u_1,\ldots,u_n) \to u$,
  $t_i = u_i$ for every $i \in 1..n$, and $t \le u$.

$methodtype(C,m)$ looks up m in C and its ancestors.

Parameter names do not affect overriding. MiniJava methods are public.

Soundness of a Type System

- A type system is *sound* if whenever $\vdash e : T$, then e will always evaluate to a value of type T in all executions.
- We want only sound rules, because then we can give a compile-time guarantee for the absence of *type errors*.
- Example of an unsound rule?

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- A type system is *sound* if whenever $\vdash e : T$, then e will always evaluate to a value of type T in all executions.
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- Example of an unsound rule?

$$\frac{\vdash e_1 : int \quad \vdash e_2 : int}{\vdash e_1 + e_2 : int[]}$$

- A programming language is called *type safe* if a compiler of the language guarantees that compiled programs will run without type errors.
  - Can be either statically or dynamically typed.

Conclusion

- Semantic analysis checks the context-sensitive properties that parsing cannot enforce.
- Symbol tables record declarations and model nested scopes; an inner declaration shadows an outer one.
- Typing judgements combine a type environment A with a class context C to check methods, statements, and expressions.
- Inheritance introduces subtyping, method lookup, overriding, and least upper bounds.

The goal is a sound guarantee: well-typed programs do not produce type errors.